



Survey Methods

3.2 - Measurement

Dr Annibale Cois, MEng, MPH, PhD
Division of Health Systems and Public Health
acois@sun.ac.za



Stellenbosch
UNIVERSITY
IYUNIVESITHI
UNIVERSITEIT

forward together
sonke siya phambili
saam vorentoe

Outline

- What does it mean “to measure”?
- Scale of measurement
- Theoretical constructs
- Operationalisation, indicators and scales
- Reliability and validity

What does it mean “to measure”?



Measurement

Measurement is the quantification of attributes of an object or event, which can be used to compare with other objects or events.

(Pedhazur and Schmelkin 2013)

Not necessarily of
'Physical' object
or experience



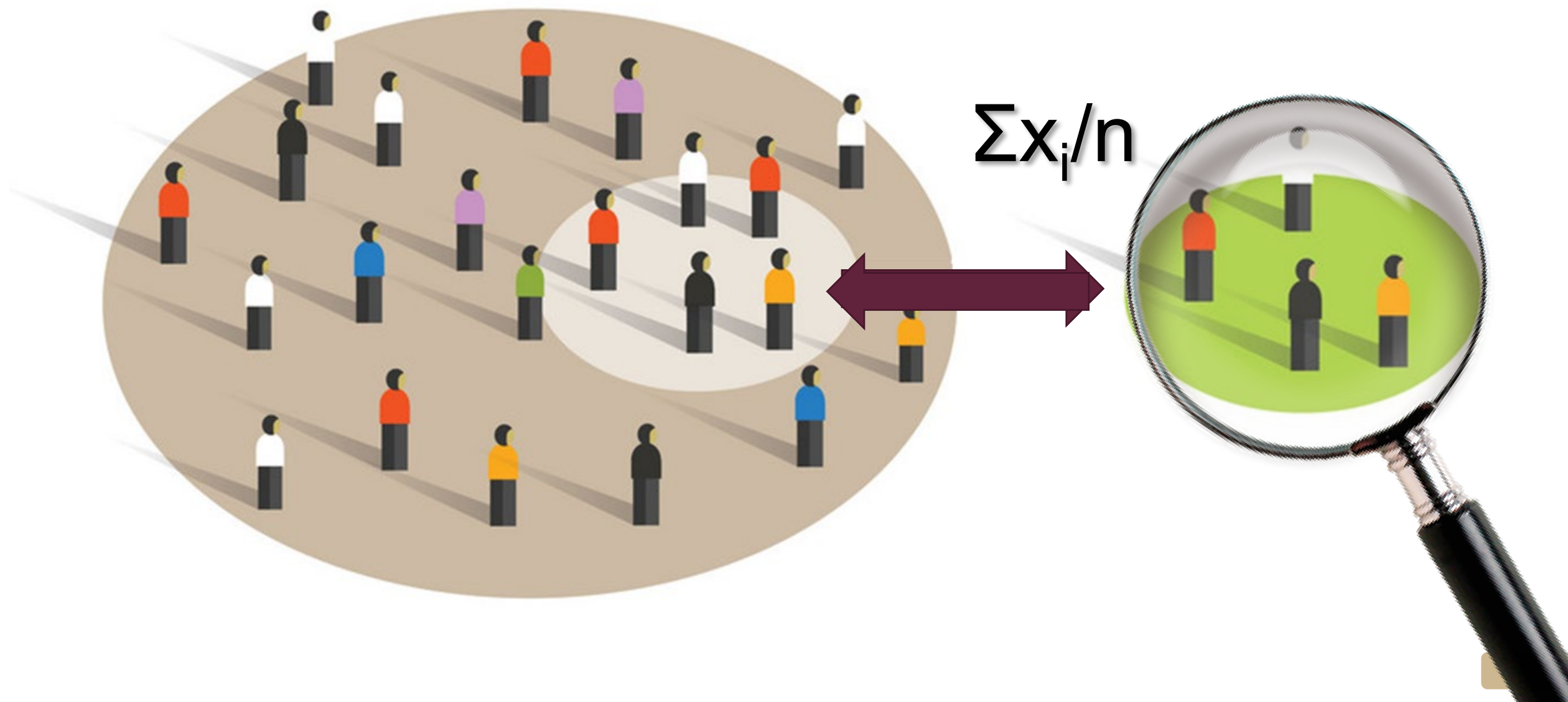
Measurement



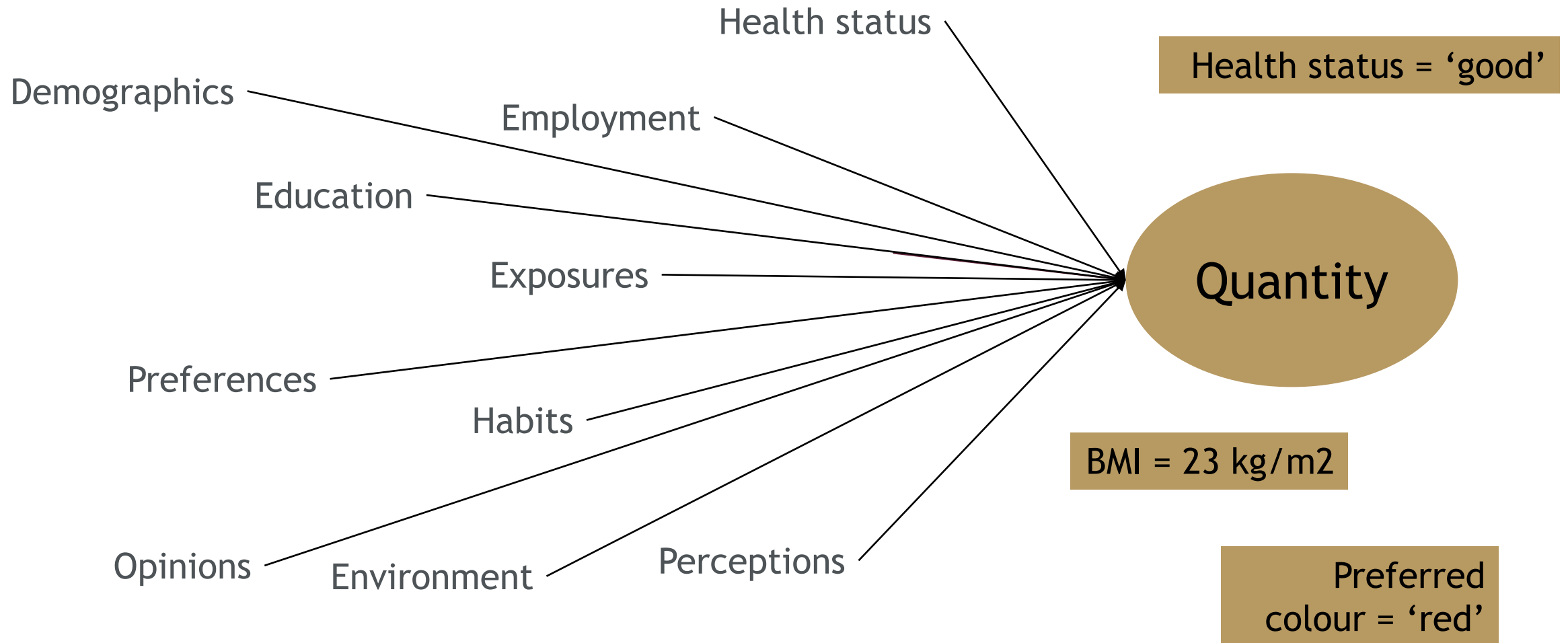
something that
can be compared
to other quantities

A "survey" is a systematic method for gathering information from (a sample of) entities for the purposes of constructing quantitative descriptors of the attributes of the larger population of which the entities are members.





What do we 'measure' in surveys?



Examples

Q1009	INTERVIEWER: Record sex of the respondent	1 MALE 2 FEMALE
-------	--	--------------------

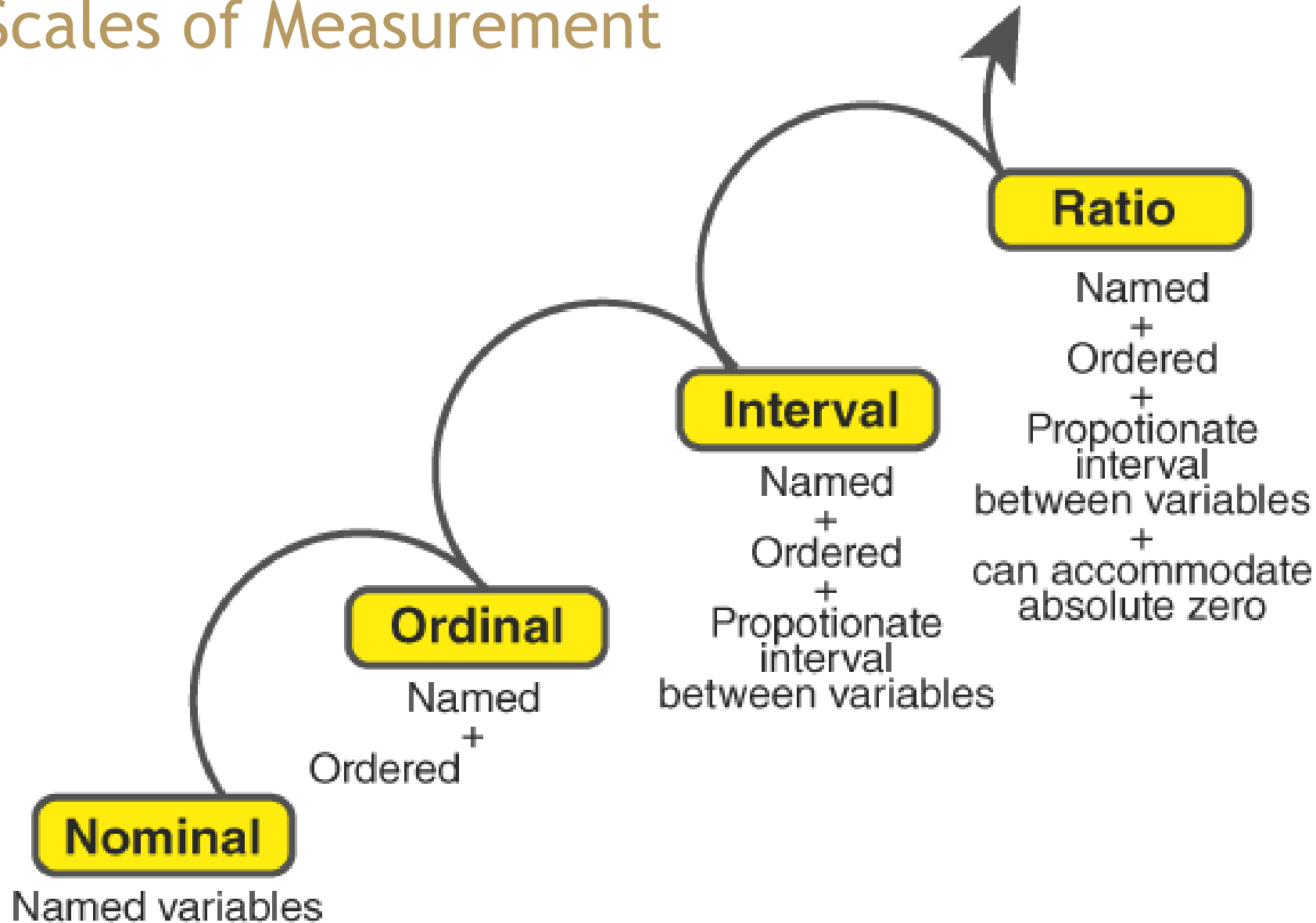
No arithmetic operation on these 'numbers'

Q2000	In general, how would you <u>rate your health</u> today?	A Very good B Good C Moderate D Bad E Very bad
-------	--	--

Q1010	What day, month and year were you born? DD / MM / YYYY Check birth certificate if available.	/ / -8 DON'T KNOW
-------	--	--

Q1014	For how many <u>years</u> have you been married or living together? INTERVIEWER: if less than 1 year, enter "00"	NUMBER OF YEARS -8 DON'T KNOW
-------	---	--

Scales of Measurement



Measurement:

Measurement is never 'trivial'...



Trends in Demographic and Health Survey data quality: an analysis of age heaping over time in 34 countries in Sub Saharan Africa between 1987 and 2015

Mark Lyons-Amos* and Tara Stones

Abstract

Objective: This paper evaluates one aspect of data quality within DHS surveys, the accuracy of age reporting as measured by age heaping. Other literature has explored this phenomenon, and this analysis build on previous work, expanding the analysis of the extent of age heaping across multiple countries, and across time.

Results: This paper makes a comparison of the magnitude of Whipple's index of age heaping across all Demographic and Health Surveys from 1986 to 2015 in Sub-Saharan Africa. A random slope multilevel model is used to evaluate the trend in the proportion of respondents within each survey rounding their age to the nearest age with terminal digit 0 or 5. The trend in the proportion of misreported ages has remained flat, in the region of 5% of respondents misreporting their age. We find that Nigeria and Ghana have demonstrated considerable improvements in age reporting quality, but that a number of countries have considerable increases in the proportion of age misreported, most notably Mali and Ethiopia with demonstrate increases in excess of 10% points.

Keywords: Data quality, Demographic and Health Survey

Measurement:

...sometimes is complex

Measuring theoretical concepts

Depression

Poverty

A construct is a **theoretical concept**, theme, or idea based on empirical observations.

It's a variable that's usually not directly measurable.

Health

Readiness

Accessibility

Acceptability

Depression

“Depression is a **mood disorder** characterized by persistent sadness and a lack of interest or pleasure in previously rewarding or enjoyable activities..”

(who, https://www.who.int/health-topics/depression#tab=tab_1)

How can we ‘measure’ depression?



CES-D-10 (Center for Epidemiologic Studies Depression Scale, short version).

Please indicate how often you have felt this way **during the past week** by using the rating scale provided.

1. I was bothered by things that usually don't bother me	0	1	2	3
2. I had trouble keeping my mind on what I was doing	0	1	2	3
3. I felt depressed	0	1	2	3
4. I felt that everything I did was an effort	0	1	2	3
5. I felt hopeful about the future	0	1	2	3
6. I felt fearful	0	1	2	3
7. My sleep was restless	0	1	2	3
8. I was happy	0	1	2	3
9. I felt lonely	0	1	2	3
10. I could not "get going"	0	1	2	3

Rarely or none of the time
(less than 1 day) → 0

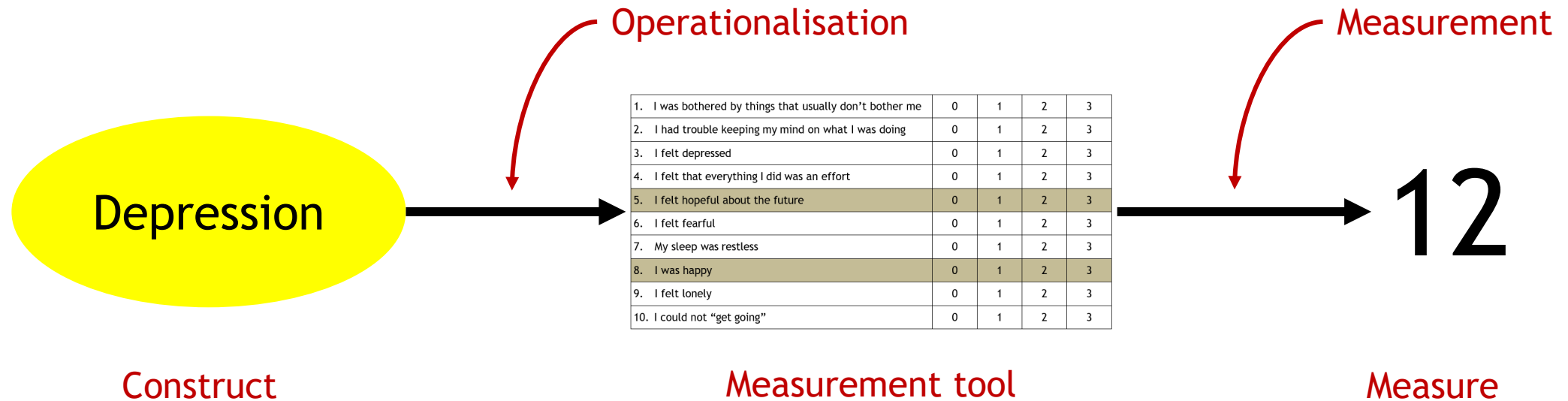
Some or a little of the time
(1-2 days) → 1

Occasionally or a
moderate amount of time
(3-4 days) → 2

Most of the time
(5-7 days) → 3

CES-D-10 (Center for Epidemiologic Studies Depression Scale, short version).

Scoring: Items 5 and 8 are reverse coded. The total score is calculated by totaling all items scored after reversing the positive mood items. Possible range for scores is 0-30 with higher scores representing greater degrees of depressed mood.



Validity

Validity concerns the extent to which a test or measure accurately assesses what it's supposed to. Validity is about the **accuracy** of a measure.

Does CESD-D-10 really
measure depression?
How do we know?

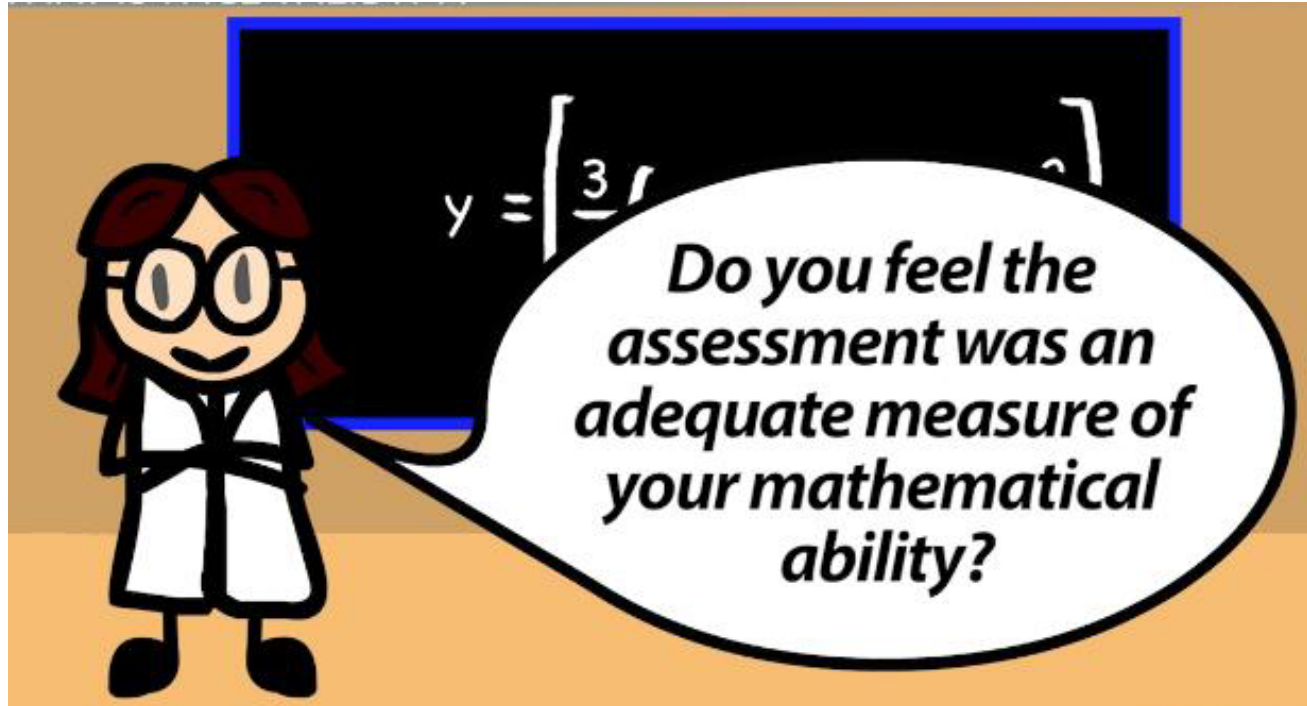


Validity

- **Face validity**: Does the content of the test appear to be suitable to its aims?
- **Content validity**: Is the test fully representative of what it aims to measure?
- **Construct validity**: Does the test measure the concept that it's intended to measure?
- **Criterion validity**: Do the results accurately measure the concrete outcome they are designed to measure?

Validity

- **Face validity**: Does the content of the test appear to be suitable to its aims?



Validity

- **Content validity:** Is the test fully representative of what it aims to measure?

“Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.”

<https://www.who.int/about/governance/constitution>



Does your self-report measurement tool only include items which ask about physical diseases?

Validity

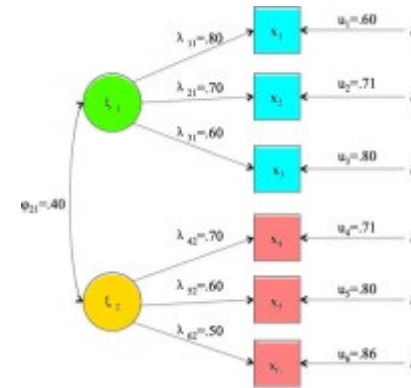
- **Construct validity**: Does the test measure the concept that it's intended to measure?



Convergent validity



Divergent validity



Factor Analysis



Predictive validity

(Confirmatory) Factor Analysis

a theory-driven approach that tests how well the data “fit” to the proposed theory.

1. I was bothered by things that usually don't bother me	0	1	2	3	
2. I had trouble keeping my mind on what I was doing	0	1	2	3	
3. I felt depressed	0	1	2	3	Positive correlation
4. I felt that everything I did was an effort	0	1	2	3	
5. I felt hopeful about the future	0	1	2	3	Positive correlation
6. I felt fearful	0	1	2	3	
7. My sleep was restless	0	1	2	3	Negative correlation
8. I was happy	0	1	2	3	
9. I felt lonely	0	1	2	3	
10. I could not “get going”	0	1	2	3	

Validity

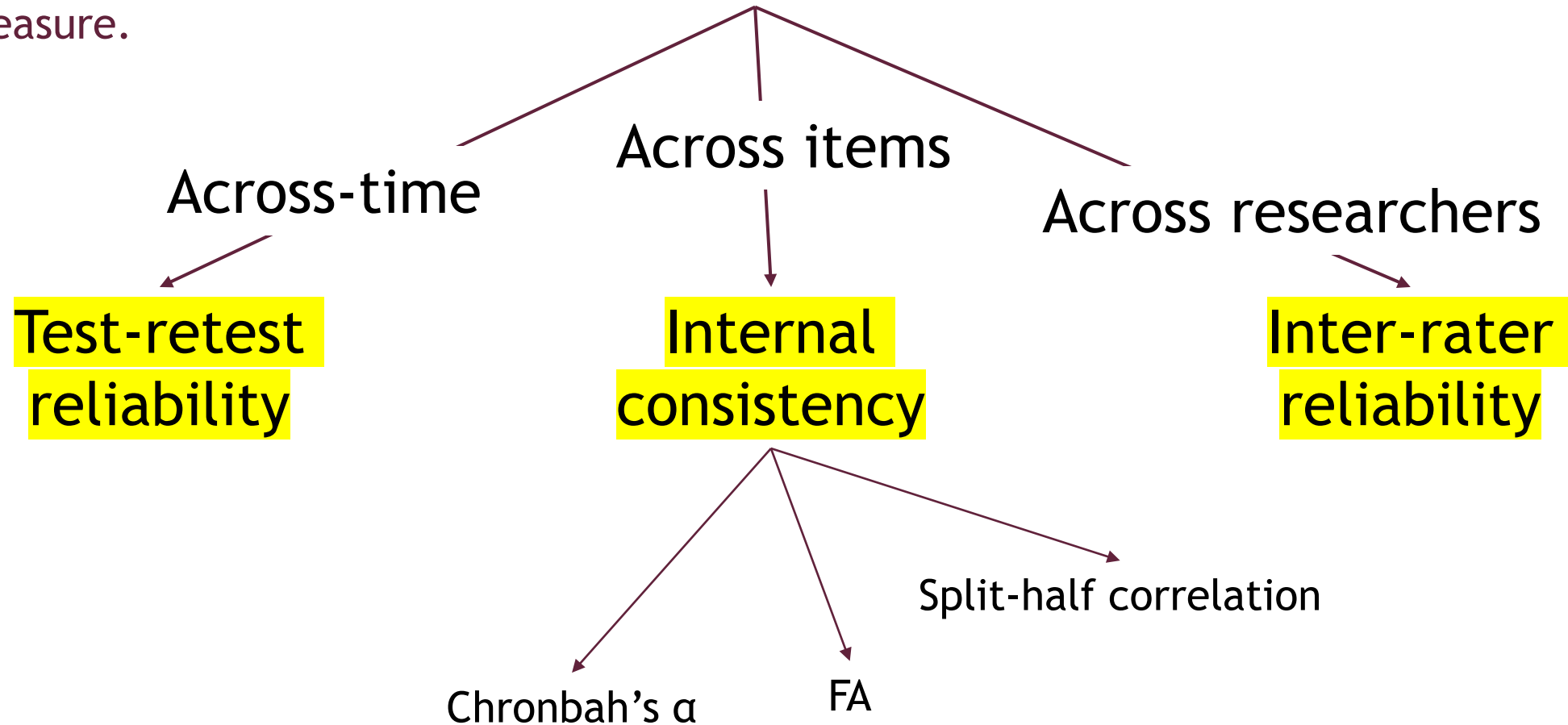
- **Criterion validity**: Do the results accurately measure the concrete outcome they are designed to measure?



Criterion validity

Reliability

The extent to which the results can be reproduced when the research is repeated under the same conditions. Reliability is about the **consistency** of a measure.



Reliability vs Validity

	Reliability	Validity
What does it tell you?	The extent to which the results can be reproduced when the research is repeated under the same conditions.	The extent to which the results really measure what they are supposed to measure.
How is it assessed?	By checking the consistency of results across time, across different observers, and across parts of the test itself.	By checking how well the results correspond to established theories and other measures of the same concept.
How do they relate?	A reliable measurement is not always valid: the results might be reproducible, but they're not necessarily correct.	A valid measurement is generally reliable: if a test produces accurate results, they should be reproducible.

RESEARCH ARTICLE

Open Access



Validation of the 10-item Centre for Epidemiological Studies Depression Scale (CES-D-10) in Zulu, Xhosa and Afrikaans populations in South Africa

Emily Claire Baron^{1*}, Thandi| Davies¹ and Crick Lund^{1,2}

- Why did the author conduct the validation?
- Which aspects did they assess?
- And how?



Abstract

Background: The 10-item Centre for Epidemiological Studies Depression Scale (CES-D-10) is a depression screening tool that has been used in the South African National Income Dynamics Study (NIDS), a national household panel study. This screening tool has not yet been validated in South Africa. This study aimed to establish the reliability and validity of the CES-D-10 in Zulu, Xhosa and Afrikaans. The CES-D-10's psychometric properties were also compared to the Patient Health Questionnaire (PHQ-9), a depression screening tool already validated in South Africa.

Methods: Stratified random samples of Xhosa, Afrikaans and Zulu-speaking participants aged 15 years or older ($N=944$) were recruited from Cape Town Metro and Ethekwini districts. Face-to-face interviews included socio-demographic questions, the CES-D-10, Patient Health Questionnaire (PHQ-9), and WHO Disability Assessment Schedule 2.0 (WHODAS). Major depression was determined using the Mini International Neuropsychiatric Interview. All instruments were translated and back-translated to English. Construct validity was examined using exploratory factor analysis with varimax rotation. Receiver Operating Characteristics (ROC) curves were used to investigate the CES-D-10 and PHQ-9's criterion validity, and compared using the DeLong method.

Results: Overall, 6.6, 18.0 and 6.9% of the Zulu, Afrikaans and Xhosa samples were diagnosed with depression, respectively. The CES-D-10 had acceptable internal consistency across samples ($\alpha = 0.69-0.89$), and adequate concurrent validity, when compared to the PHQ-9 and WHODAS. The CES-D-10 area under the Receiver Operator Characteristic curve was good to excellent: 0.81 (95% CI 0.71–0.90) for Zulu, 0.93 (95% CI 0.90–0.96) for Afrikaans, and 0.94 (95% CI 0.89–0.99) for Xhosa. A cut-off of 12, 11 and 13 for Zulu, Afrikaans and Xhosa, respectively, generated the most balanced sensitivity, specificity and positive predictive value (Zulu: 71.4, 72.6% and 16.1%; Afrikaans: 84.6%, 84.0%, 53.7%; Xhosa: 81.0%, 95.0%, 54.8%). These were slightly higher than those generated for the PHQ-9. The CES-D-10 and PHQ-9 otherwise performed similarly across samples.

Conclusions: The CES-D-10 is a valid, reliable screening tool for depression in Zulu, Xhosa and coloured Afrikaans populations.

Keywords: Validation, Depression, CES-D-10, South Africa, Prevalence, PHQ-9, Screening

Summary questions

- Concept of measurement
- Scales of measurement
- What is construct?
- Operationalisation, indicators and scales
- What are reliability and validity? How do you measure them?

References

Baron, Emily Claire, Thandi Davies, and Crick Lund. 2017. 'Validation of the 10-Item Centre for Epidemiological Studies Depression Scale (CES-D-10) in Zulu, Xhosa and Afrikaans Populations in South Africa'. *BMC Psychiatry* 17 (1): 6.

<https://doi.org/10.1186/s12888-016-1178-x>.

Chiang, I.-Chant A., Rajiv S. Jhangiani, and Paul C. Price. 2015. 'Reliability and Validity of Measurement', October.

<https://opentextbc.ca/researchmethods/chapter/reliability-and-validity-of-measurement/>.

Other resources

Lyons-Amos, Mark, and Tara Stones. 2017. 'Trends in Demographic and Health Survey Data Quality: An Analysis of Age Heaping over Time in 34 Countries in Sub Saharan Africa between 1987 and 2015'. *BMC Research Notes* 10 (1): 760. <https://doi.org/10.1186/s13104-017-3091-x>.

Pedhazur, Elazar J, and Liora Pedhazur Schmelkin. 2013. *Measurement, Design, and Analysis: An Integrated Approach*. psychology press.

Tavakol, Mohsen, and Angela Wetzel. 2020. 'Factor Analysis: A Means for Theory and Instrument Development in Support of Construct Validity'. *International Journal of Medical Education* 11 (November): 245–47.

<https://doi.org/10.5116/ijme.5f96.0f4a>.



Thank you

Annibale Cois

Education Building, 4th floor, Room 4060B

acois@sun.ac.za